NFL visual data analysis application

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1 Motivation

In the highly competitive world of the National Football League (NFL), the difference between win and loss is marginally small. Player performance, strategic tactics, and game dynamics are all influenced by multiple factors, including player movements. Analyzing these movements can provide insights into player behavior, team tactics, and defensive as well as offensive schemes. Therefore, we present a tool for analyzing NFL data visually, in this paper.
The application is built to serve coaches, players, analysts, and enthusiasts by providing a dynamic and interactive platform for visualizing game data. It offers a detailed representation of player positions, movements, and play outcomes, making it a valuable tool for tactical analysis and decision-making.

The core functionality of the application includes:

1. **Visualization of Player Movements**: The application graphically displays the players positions over the time at a given play on the field, allowing users to trace the flow of the game.
2. **Interactive Play Analysis**: Users can select specific games, downs, play types (run or pass), plays and more to analyze. The application provides a breakdown of the plays, the involved players and the outcome of each play.
3. **Detailed Statistical Insights**: The application offers statistical summaries, such as distance to the end zone, yard line, result of the play, and offensive formations.
4. **Speed and Acceleration Analysis**: The application visualizes the speed and acceleration of players over time, providing insights into players performance and physical conditioning.
5. **Interaction Heatmap**: The application generates heatmaps based on the proximity of players to each other and the football, highlighting areas of high interaction and potential strategic points on the field.

Through this application, we aim to demonstrate the power of data visualization in sports and to make the analyzing process for coaches and players more structured and easier to handle.

2 Data Preprocessing

The used data is provided by The National Football League via the website Kaggle and needs preprocessing in order to be integrated into the graphical user interface. Since some columns in each data set are not being used, they are removed when the data are loaded for the first time. The used data sets are weekly data for the weeks one to seventeen, games data, and plays data. The "week" data sets include information regarding timestamps, coordinates of players and the football, NFL, game and play ID, jersey numbers, positions, and player names. The "games" data set contains the game ID, the date, the week, and both team names. Lastly, the "plays" data set includes the game and play ID, a play description, and several other information about each play. To simplify usage, all 17 data sets containing information on weekly gamedays are combined into one data frame. The next step involves identifying and removing any missing or duplicated values from the data sets, which proved to be negligible since almost all of those values can be found in the weeks data set, where the football is listed the same as any other player but does not have the same attributes as a normal player (e.g. jersey number).

3 Design

Following the successful completion of the preprocessing stage, we proceeded to the graphical user interface (GUI) of the application. The initial decision point was the selection of the appropriate programming language and associated packages. Our team members were the most advanced in Python, so we decided to implement the application with this language. With regard to the packages, we considered two options: Tkinther or dash. We concluded that dash, an open-source package, was a more suitable choice for the interactive design we had in mind. Dash is a web application framework that allows for the implementation of a range of customisations, making it an ideal choice for the creation of interactive web applications.

Design Process

The development of the football analysis software was guided by principles from key texts in interaction design and human-computer interaction, namely [CRC13], and [SPC+18]. The following paragraph details the design process, focusing on User-Centered Design (UCD) and Interaction Design, ensuring that the software meets user needs and provides an engaging, intuitive experience.

The primary aim of using user-centred design (UCD) was to ensure that the software met the specific needs and preferences of its end users, which includes coaches, analysts and players. By focusing on the users goals and tasks, the design process aimed to create a product that was both effective and easy to use. The process began with an analysis of existing football analysis software. Next,

we leveraged on the expertise of our domain experts, Robert and Lukas. Their practical knowledge and experience provided valuable insights into the specific needs and workflows of coaches, analysts and players. The idea of user-centred design can also be found in Shneiderman's task by data type taxonomy, which aims to categorise and design visualisations based on user tasks.

Interaction design principles are crucial for enhancing user engagement and usability. In terms of the user interface (UI) design, an intuitive layout with filters on the left side was created for easy access and interaction. This design choice tried to ensure clarity and simplicity, making the interface easy to navigate. The design followed Shneiderman's "overview first, zoom and filter, details-on-demand" strategy". This approach provides users with a broad overview of the data, displayed during the filtering (in infoboxes), before showing the plots. The more detailed view is shown by selcting "show". This way the application is only updated when necessary.

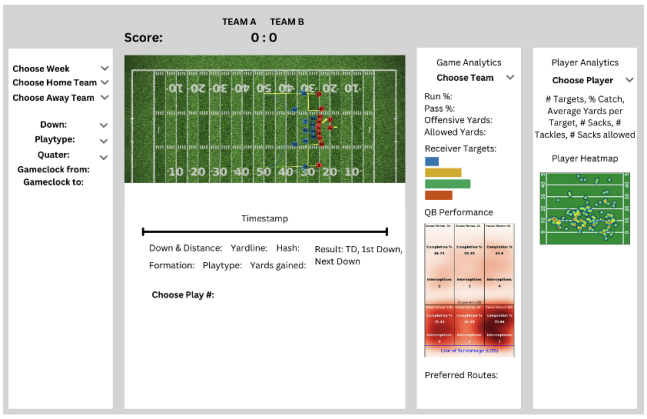


Fig. 1. Prototype Football Analysis Application

The initial design concept presented in the first presentation served as the basis for the subsequent implementation. It is important to note that the feedback received during the first presentation has been taken into account and the focus has shifted to the visualisation of the playing field and game information to ensure the creation of a working prototype. Furthermore, we drew inspiration in visualising the plot from [BS18] and [Sil20]

As shown in the drawn prototype (Figure 1), the contents of the application include the following:

- Filters
- Score
- Information regarding the play (two info boxes)

- Football field
- Slider for the visualisation

The filters include the following: home team, away team, game, play, week, down and quarter. In our opinion, these were the most necessary filters to retain the play. All filters were created to be interactive with each other. The idea is that initially, the home team X is chosen, and the following filter (game) only shows all away teams the team played against. Finally, a specific play must be selected. This is essential to ensure the stability and usability of the application. Each game comprises approximately 100 plays, with each play comprising approximately 80 frames, which are necessary for visualising the movement of the plays. For the filters, we have opted for a drop-down menu as the most suitable option for an intuitive selection process. Regarding the choice of different plays, the decision was between a drop-down menu and a slider.

The purpose of this visualisation is to improve the understanding and analysis of individual plays by animating the movements of players on a football field. The football field is implemented by using the X and Y coordinates of the frames corresponding to the selected play in the filters. First, a function "animate play" is defined which takes the parameters gameId, playId, plays and weeks. The plays DataFrame contains metadata about the plays, while weeks contains detailed tracking data for each play. Within the function, the data is filtered to isolate the specific game and play specified by gameId and playId. The frames of X and Y data represent a snapshot of the play, which is necessary to visualise the movement. Within each frame, player positions are plotted, with players differentiated by team (home or away) and the football itself. Each frame is appended to the list of frames and a corresponding step is added to the slider to indicate the frame's position in the animation. Players' positions are indicated by markers and their shirt numbers are displayed on these markers. Different colours are used to distinguish the teams: blue for the away team, red for the home team and brown for the football. The layout of the field includes several key elements, including the playing field, the line of scrimmage (red dashed line), the first down marker (yellow dashed line and annotation of the down number), the field lines (white lines at regular intervals to represent the yard lines), the field line numbers (numbers on the field lines going from one end zone to 50 yards in the middle to 10 yards again at the other end zone at 10 yard intervals), the end zones (light purple, almost grey), with notes for the home and visiting teams). Additional textual information, such as the down and the game clock, is displayed at the top of the plot to ensure the user knows which play is selected. The final layout adjustments involve configuring the legend, which displays player information (name and jersey numbers). The start and stop buttons in the animate play function are implemented using Plotly's updatemenus feature. These buttons provide users with control over the animation of the football play, allowing them to play or pause the sequence of frames that depict the movements of players on the field. The plot can be seen in the Figure 2.

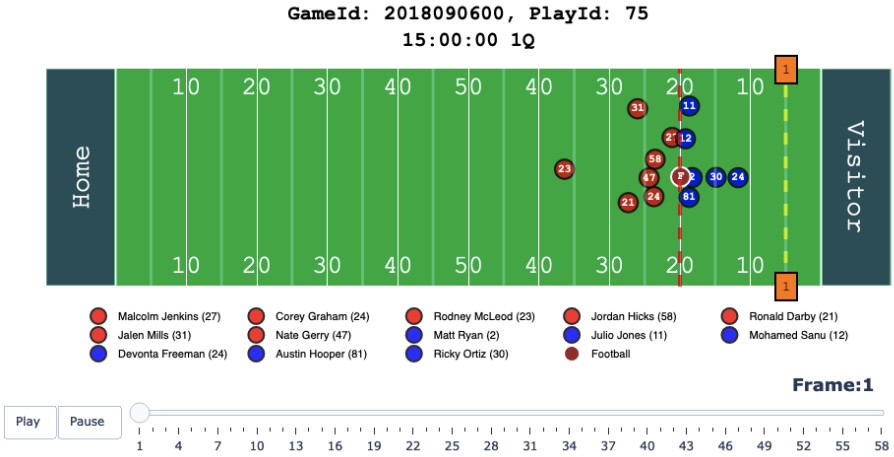


Fig. 2. Football Field Plot

In addition to visualising the field and the movement of the players, we wanted to provide the user with additional information. The first and most obvious way to do this was to include a text box with information about the match. On the left hand side of the application, below the filters, we included information about the match. This should give the user an initial insight into the game they are viewing. It is important to note that this information is displayed before the plots are updated with the Show button. The idea is that we only update the plots if the user has selected the correct play. The filters and information box are shown in the figure (3) below.

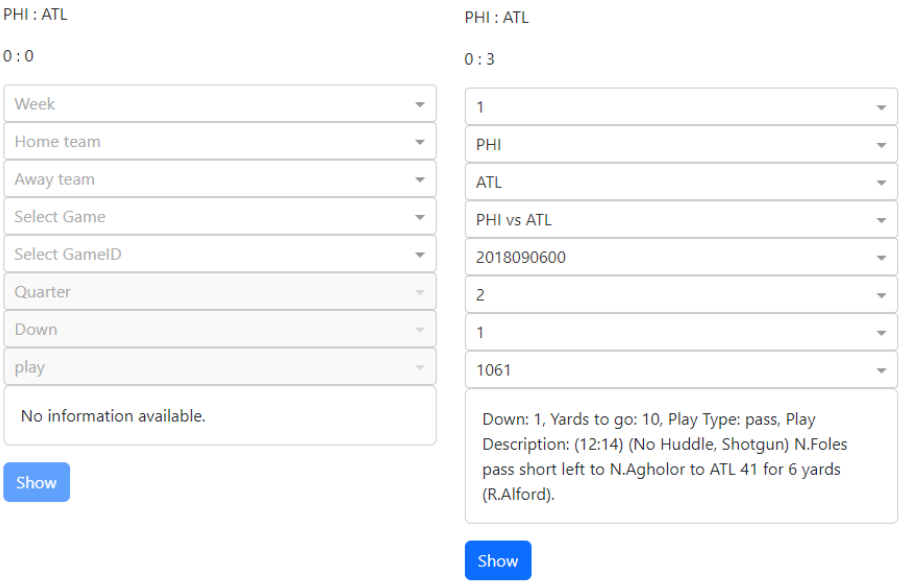


Fig. 3. Filters and Information Box

Furthermore, we have designed two distinct plots to enhance the visualization of player-related data.

1. **Player acceleration and speed profile**: The graph shows a player's speed and acceleration over time for the selected play. The graph itself is interactive, with a drop-down menu to select a player. Coaches and analysts can identify key moments such as the burst of speed or the slowdown of a player. It enhances the visualisation of the football field by providing player centric information. (Figure 4)
2. **Player proximity heatmap**: The graph shows proximity to other players and to the football in x and y coordinates. It allows analysts to examine how player interactions affect the outcome of plays. For example, it can show how well a defensive line stays together or how effectively a receiver shakes off defenders. The visualisation has a proximity threshold of 20 yards to exclude outliers and improve the readability. The assumption is that players who are that far apart are not of interest. (Figure 5)

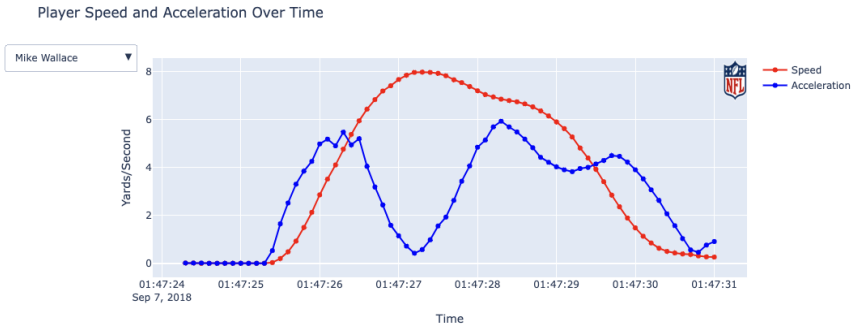


Fig. 4. Speed and Acceleration

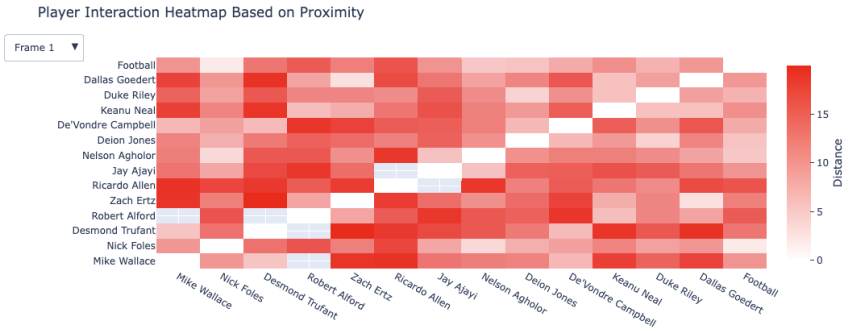


Fig. 5. Proximity of players

4 Discussion

4.1 Advantages and Disadvantages of Our Approach

Our approach to developing the American Football Analysis tool has several notable advantages. Firstly, the tool provides a highly functional and interactive platform for visual analytics, allowing users to explore the data in a dynamic and engaging manner. Our visual tools allow users to achieve a deeper understanding of the data. Furthermore, the tool is designed to be practical for real-life applications, offering valuable insights that can be utilized by coaches, analysts, and players to enhance their performance and strategies.

However, our approach also has some disadvantages. One significant challenge is the high code complexity associated with developing such an advanced tool. Our usage of dash components and their often unwanted callback interactions resulted in rather unstructured and complex code. Although it's functionality is out of question, a refactoring of most of the functions is necessary to ensure a good code base for further additions and implementations.

4.2 What Went Well?

Several aspects of the project were particularly successful. Effective team collaboration played a crucial role in our success. Each team member contributed their unique skills and perspectives, leading to a well-rounded and thoroughly developed tool. Additionally, some of our team members had some real-life experience in American Football, which allowed us to incorporate practical insights and ensure that our tool was relevant and useful for real-world applications.

4.3 What Were the Difficulties?

Despite our successes, we encountered several difficulties during the development process. Managing large, non-continuous datasets posed a significant challenge. The complexity of the data required careful handling to ensure that our tool could process and visualize it efficiently. Additionally, ensuring complex interactivity within the tool was challenging, especially in the filtering section. The integrated callback functions of Dash often introduced unexpected behaviors and performance bottlenecks, requiring extensive debugging and optimization to achieve the desired user experience.

4.4 What Would You Add If You Had More Time?

Given more time, we would incorporate several additional features to enhance the functionality and user experience of the tool. One potential addition is the incorporation of player heatmaps. This feature would provide a visual representation of player movements and positions over the whole game, offering more specific insights of a players performance. We would also implement lazy loading

to improve the tool's performance. Since the data we worked with consisted of 20 dataset with more than 100 Megabytes each, the initial loading times were unsustainable. Lastly, enhancing the graphics of the tool would provide a more polished and professional user interface, making the tool more appealing and easier to use.

4.5 Use Cases

The American Football Analysis tool has several practical use cases that demonstrate its value:

- **Coaches and Analysts:** Coaches can use the tool to analyze player performance, identify strengths and weaknesses, and develop strategies based on data-driven insights. Analysts can delve into detailed game data to uncover patterns and trends that can inform future decisions.
- **Players:** Players can utilize the tool to review their own performance metrics, understand their positioning and movements during games, and receive feedback to improve their skills and tactics.
- **Scouts and Recruiters:** Scouts and recruiters can leverage the tool to evaluate potential recruits by analyzing their game data, comparing player statistics, and identifying promising talent based on objective performance data.
- **Fans and Enthusiasts:** Although the tool is primarily made for coaches, scouts and players, fans and enthusiasts who wish to engage with game data on a deeper level can also use this model.

5 Conclusion

The NFL visual data analysis application presented in this paper is a powerful tool designed to support coaches, players and football interested parties analysing previous games and use the obtained insights to, i.e. develop sensible game plans or getting insights of specific players movements. By providing dynamic and interactive visualizations of player movements, play outcomes, and detailed statistical insights, the application bridges the gap between raw data and meaningful insights.

The core functionalities of the application, including the visualization of the player movements, interactive play analysis and statistical summaries, offer a detailed view into the dynamics of a football game. The addition of the speed and acceleration analysis, as well as the interaction heatmaps, offer especially players and athletic coaches deep insights into the performances of each player during games. These features make it easier for users to understand player behavior, team tactics and the overall flow of the game, providing the users necessary information to develop game plans.

The development process, guided by principles of user-centered design and interaction design, ensured that the application is both intuitive and effective. By using python as a programming language and Dash as framework, we were able to create an interactive interface that ensures simple and intuitive use for the user.

Through this application, we aim to demonstrate the potential of data visualization in sports analysis. The ability to visualize large and complex datasets in a meaningful way could enhance the elaboration of game plans in the game of football. Furthermore, the design and functionality provides the foundation to enhance the application in the future. For example developing and elaborate real-time data analysis, as well as additional statistical displays.

In conclusion, the NFL visual data analysis application is a contribution to the area of sports analytics, providing useful tools for understanding and analysing strategies in football games.

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